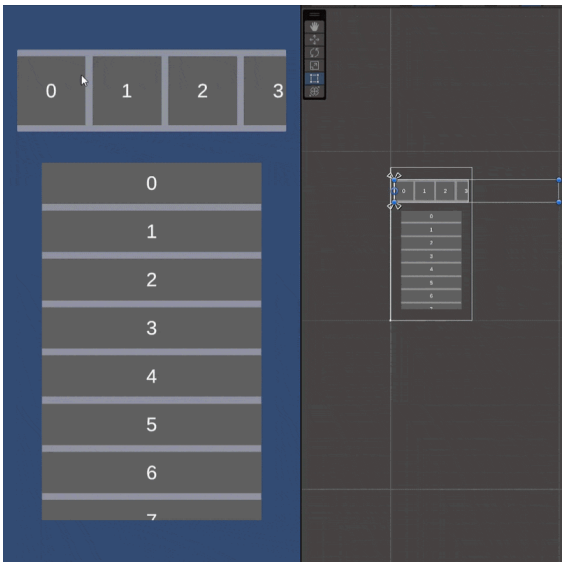
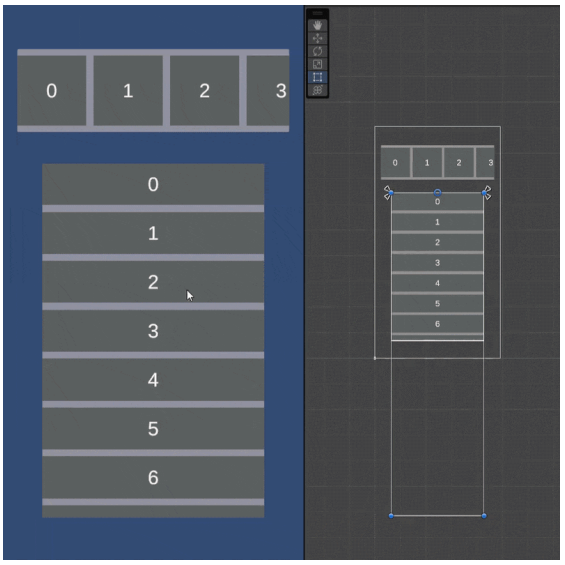
Recyclable Scroll - Reusing UI Elements

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https://github.com/son24tuoi/recyclable-scroll-unity.git





The scroll system optimizes performance by reusing UI elements (GameObjects). Instead of creating thousands of objects that consume RAM and slow down CPU processing, this system maintains only a small number of elements sufficient to display on the screen and updates their data as the screen is scrolled.

Key Features

- High Performance: Only creates the minimum number of objects necessary to fill the viewport.
- Support for large datasets: Can display lists of up to thousands of items without increasing resource consumption.
- Easy integration: Based on Unity UI's standard ScrollRect, the source code can be customized to suit individual preferences.

Comparison with Traditional Methods

Criteria	Traditional Scroll	Reusable Scroll
Number of Objects	Equal to the total amount of data → the more data, the more objects.	Fixed, typically twice the number visible in the viewport.
RAM usage	Increases gradually with data	Low and stable

Operating principle

- Pagination: Uses the `pageIndex` variable to control data boundaries.
- Scrolling: When *content* reaches the edge of the *viewport*, it is moved to the appropriate position to allow further scrolling (while updating the displayed data).

Quick Setup Guide

1. Attach the `RecyclableScrollVertical.cs` / `RecyclableScrollHorizontal.cs` script to the *GameObject* containing the *ScrollRect* (you can place it elsewhere too 😅).
2. Drag the `ScrollRect` and `ContentRT` variables for `RecyclableScrollVertical.cs` / `RecyclableScrollHorizontal.cs` onto the Inspector window. Set the maximum number of list items to display in `TotalItems` (or set it via code). Other variables visible in the Inspector will update automatically.
3. Configure the UI items in the list as child objects of the *Content* game object (make sure the number is just enough to fill the viewport).
4. Configure `RecyclableScrollVertical.cs` / `RecyclableScrollHorizontal.cs` using the `Init()` function with a callback function as a parameter to update the data.

See the sample script `ItemLoader.cs` and the sample scene *Scene_RecyclableScroll* in *Samples*.

5. Retrieve the list of UI items in *the Content* using the `TryGetComponentsInContentChildren()` function. See the sample script `ItemLoader.cs` and the sample scene *Scene_RecyclableScroll* in *Samples*.